CSCI323 Modern Artificial Intelligence Project Description

Executive Summary

This document provides a comprehensive project description for the CSCI323 Modern Artificial Intelligence course at the University of Wollongong Dubai (UOWD). The project requires students to identify a real-world business problem within a UAE-based company and develop an AI-powered solution using appropriate techniques and programming languages. The project emphasizes practical implementation, demonstrating theoretical knowledge through industry-relevant applications, and critical evaluation of solutions in authentic business contexts.

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1. Project Overview and Objectives

1.1 Course Context

The CSCI323 Modern Artificial Intelligence project is designed to bridge the gap between theoretical concepts taught in lectures and practical implementation challenges faced by organizations. Students will apply AI methodologies to solve genuine business problems, gaining hands-on experience with industry-standard tools and techniques.

Course Instructors: Dr. Milan Dordevic, Dr. Abdullah El Nokiti, Ms. Asma Damankesh, Ms. Syama Kurungodathil
Academic Year: Spring 2026
Institution: University of Wollongong Dubai (UOWD)

1.2 Learning Outcomes

Upon successful completion of this project, students will be able to:

- Identify business problems suitable for AI-based solutions within UAE organizations
- Evaluate the appropriateness of different AI techniques for specific use cases
- Implement AI solutions using programming languages (Python, C++, C#, R, Java, or similar)
- Conduct rigorous data analysis and validate solution effectiveness
- Communicate technical findings to both technical and non-technical stakeholders
- Reflect critically on project outcomes and lessons learned from industry engagement

1.3 Project Goals

1. **Practical Application:** Demonstrate mastery of AI concepts through real-world problem-solving
2. **Industry Engagement:** Build direct relationships with UAE-based companies and understand their operational challenges
3. **Solution Development:** Create fully functional, documented, and tested AI solutions
4. **Professional Communication:** Present findings in formats suitable for business and academic audiences

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2. Project Scope and Requirements

2.1 Company Selection

Students must identify and partner with a **real company located in the UAE**. This can be achieved through:

- **Direct Industry Engagement:** Contacting companies directly to discuss their operational challenges
- **Research-Based Approach:** Conducting secondary research on generic industry problems that typically require AI solutions
- **Domain Focus:** Selecting companies from diverse sectors (retail, manufacturing, finance, healthcare, logistics, hospitality, etc.)

2.2 Problem Identification and Scoping

The identified problem must:

- Be **specific and well-defined** within the selected company's operational context
- Be **solvable using AI techniques** discussed in CSCI323 (machine learning, deep learning, reinforcement learning, etc.)
- Have **measurable business impact** (cost reduction, efficiency improvement, customer satisfaction, revenue growth, risk mitigation)
- Be **achievable within project timeline and resource constraints**

2.3 Allowed Programming Languages and Technologies

Students may use:

- **Python** (recommended for most AI applications)
- **C++, C#**
- **R** (statistical and data analysis focus)

- **Java**
- **Any other approved language**

Students should select the language and frameworks most appropriate for their specific problem and solution design.

2.4 Project Team Structure

- **Team Size:** Groups of 5 students
- **Role Distribution:** Clear assignment of responsibilities among team members
- **Collaboration:** Evidence of effective teamwork, code integration, and mutual learning

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3. Core Project Components (Three Main Aspects)

3.1 Aspect 1: Motivation and Problem Definition

Objective: Clearly articulate the business problem and justify why AI is the appropriate solution.

Key Elements:

- **Problem Statement:** Detailed description of the business challenge or opportunity the company faces
- **Business Context:** Industry background, competitive landscape, and operational context
- **Current Approach:** How the company currently handles the problem and its limitations
- **Why AI?** Explanation of why traditional or manual approaches are insufficient
- **Expected Benefits:** Quantifiable and qualitative benefits the AI solution will deliver
- **Stakeholder Engagement:** Evidence of understanding company stakeholder needs and constraints

Deliverables:

- 1–2 pages of narrative motivation
- Business impact assessment and success metrics
- Company profile and operational context overview

3.2 Aspect 2: Solution Implementation

Objective: Demonstrate technical expertise in selecting, implementing, and deploying an AI solution.

Key Elements:

2.1 Technical Requirements Analysis

- System requirements (data, computational resources, infrastructure)
- Integration with existing company systems
- Scalability and performance considerations
- Security and privacy requirements

2.2 Methodology and Approach

- AI techniques selected (supervised learning, unsupervised learning, reinforcement learning, deep learning)
- Justification for technique selection based on problem characteristics
- Literature review and relevant research
- Comparison of alternative approaches considered

2.3 Data and Dataset

- Data source, collection, and availability (real or synthetic, with justification)
- Data preprocessing, cleaning, and feature engineering
- Dataset characteristics (size, distribution, quality issues)
- Train/validation/test split strategy

2.4 Implementation Details

- Architecture design and system diagrams
- Algorithm implementation with code walkthroughs
- Model training, hyperparameter tuning, and validation
- Development tools and libraries used
- Version control and collaborative development approach

2.5 Validation and Testing

- Evaluation metrics appropriate to the problem (accuracy, precision, recall, F1-score, RMSE, AUC, business KPIs, etc.)
- Test strategy and test cases
- Comparison against baselines or existing solutions
- Error analysis and failure case identification

Deliverables:

- 3–4 pages of technical methodology
- Screenshots of implementation, data processing, and model training
- Code repository with clean, well-commented code
- Detailed algorithm explanation and pseudocode
- Performance evaluation results and comparisons
- System architecture and workflow diagrams

3.3 Aspect 3: Solution Evaluation and Lessons Learned

Objective: Critically evaluate the solution's effectiveness and reflect on practical and theoretical learning.

Key Elements:

3.1 Solution Effectiveness

- Extent to which the solution addresses the original business problem
- Quantitative results: model performance metrics and business impact
- Qualitative assessment: feedback from company stakeholders (if available)
- Comparison against objectives and success criteria established in Aspect 1

3.2 Practical Challenges and Constraints

- Technical challenges encountered and how they were resolved
- Limitations of the solution (algorithmic, data-related, computational)
- Resource constraints and trade-offs made
- Scalability barriers and mitigation strategies
- Ethical and compliance considerations (data privacy, bias, fairness)

3.3 Recommendations for Implementation

- Proposed deployment strategy for the company
- Ongoing maintenance and model monitoring plan
- Future enhancements and extensions
- Integration into company operations and workflows

3.4 Lessons Learned

- Key insights gained from theory-to-practice experience
- How theoretical concepts from the course manifested in real-world implementation

- Gaps between academic theory and industry practice
- Personal and team learning outcomes
- What you would do differently if you could restart the project

3.5 Business and Societal Impact

- Quantified or estimated business value (cost savings, revenue increase, risk reduction)
- Broader implications for the industry or society
- Ethical considerations and responsible AI practices employed
- Scalability to other companies or problem domains

Deliverables:

- 2–3 pages of critical evaluation and reflection
- Limitations and constraints discussion
- Recommendations for company adoption
- Lessons learned summary
- Future work and enhancement roadmap

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4. Report Structure and Deliverables

4.1 Written Project Report

The final project report must follow this structure:

1. **Title Page**
    - Project title, team members, student IDs, submission date, instructor name
2. **Executive Summary** (½–1 page)
    - High-level overview of problem, solution, and key findings
    - Business impact and recommendations
    - Suitable for non-technical stakeholders
3. **Table of Contents and List of Figures/Tables**
4. **Introduction and Motivation** (1–2 pages)
    - Company and industry background
    - Problem statement and significance
    - Why AI is the appropriate solution
    - Business objectives and success metrics
5. **Literature Review** (1–2 pages)

- Relevant research and industry practices
- Comparison of AI approaches and techniques
- Theoretical foundation for selected methodology

6. **Technical Requirements and Analysis** (1 page)
    - System requirements and constraints
    - Data requirements and availability
    - Integration and scalability considerations
7. **Solution Methodology and Implementation** (3–4 pages)
    - AI technique(s) selected and justified
    - Data preprocessing and feature engineering
    - Algorithm design and implementation architecture
    - Model training, tuning, and validation procedures
    - Screenshots and code samples
8. **Results and Evaluation** (2–3 pages)
    - Model performance metrics and results
    - Visual representations (graphs, charts, confusion matrices)
    - Comparison against baselines or alternatives
    - Business impact assessment
9. **Limitations, Challenges, and Constraints** (1 page)
    - Technical and data-related limitations
    - Resource and time constraints
    - Scalability and generalization challenges
    - Ethical and compliance considerations
10. **Conclusions and Discussion** (1–2 pages)
    - Summary of findings and contributions
    - Achievement of objectives
    - Key lessons learned
    - Insights gained from industry engagement
11. **Recommendations and Future Work** (1 page)
    - Recommendations for company implementation
    - Proposed enhancements and extensions
    - Maintenance and monitoring strategy
    - Broader applications and scalability
12. **References** (APA or IEEE format)

- Academic papers, textbooks, online resources
- Company documentation and stakeholder communications
- Software tools and frameworks documentation

13. **Appendices**
    - Full code listings and algorithms
    - Detailed data preprocessing procedures
    - Additional evaluation metrics and charts
    - Interview summaries or company correspondence

14. Demo

- Video recording of your project

4.2 Presentation Deck

A professional presentation (10–15 slides) summarizing the project:

- **Slide 1:** Title, team members, institution, date
- **Slides 2–3:** Problem motivation and business context
- **Slides 4–5:** Solution approach and methodology
- **Slides 6–8:** Implementation, architecture, and key results
- **Slides 9–10:** Evaluation metrics, performance, and business impact
- **Slides 11–12:** Limitations and lessons learned
- **Slides 13–14:** Recommendations and future work
- **Slide 15:** Q&A and contact information

Presentation Guidelines:

- Clear, concise language suitable for mixed audience (technical and non-technical)
- Visual elements: diagrams, screenshots, charts, and graphs
- Focus on key findings and business value
- Avoid dense text; use bullet points and visuals
- Practice and timing: 12–15 minutes for presentation + Q&A

4.3 Code Repository and Codebase

Students must provide:

- **Source Code:** Well-organized, commented, and documented code
- **Version Control:** GitHub, GitLab, or similar (with clear commit history)
- **Documentation:** README file with setup instructions, dependencies, and usage guide
- **Data:** Sample dataset (or instructions for obtaining actual data if proprietary)

- **Reproducibility:** All steps to reproduce results from data to final model
- **Deliverables:** All code files compressed in a .zip folder or provided via repository link

Code Quality Standards:

- Clear, meaningful variable and function names
- Comments explaining complex logic and design decisions
- Modular structure and separation of concerns
- No hardcoded paths or environment-specific configurations
- Testing code or unit tests (where applicable)

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5. Assessment and Grading Rubric

Projects will be evaluated on a 100-point scale using the following criteria:

Criteria	Excellent (85–100)	Good (70–84)	Satisfactory (50–69)	Needs Improvement (<50)	Weight
Quality of Project Design	Rigorous, innovative, well-executed design addressing real business problem	Clear and mostly sound project scope and methodology	Adequate design, lacks some depth or innovation	Unclear or poorly defined project scope	30%
Data Analysis and Implementation	Detailed, correct, and appropriate analytical methods; thorough implementation	Mostly correct analysis with minor issues; solid implementation	Basic analysis or partial implementation; functional but basic	Incomplete or flawed analysis; non-functional implementation	30%
Structure, Style, and Referencing	Professional layout, coherent writing, proper citations throughout	Mostly well-written, well-organized, proper referencing	Acceptable structure and tone; some citation issues	Poor organization, inconsistent style, inadequate referencing	20%

Findings, Discussion, and Impact	Thoughtful insights, strong conclusions, clear business impact	Relevant results with interpretation; meaningful discussion	Basic discussion of findings; limited impact assessment	Weak findings, vague conclusions, minimal discussion	20%

Additional Considerations:

- **Presentation Quality:** Clarity, engagement, time management, and ability to handle questions
- **Peer Evaluation:** Team member contribution and collaboration (peer-assessed)
- **Code Quality:** Organization, documentation, reproducibility, and best practices

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Submission Format:

- Report: PDF or Word document
- Presentation: PDF or PowerPoint
- Code: GitHub/GitLab repository link or .zip file
- All submissions via learning management system (LMS) unless otherwise specified

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6. Guidelines and Best Practices

6.1 Project Management

- **Regular Meetings:** Weekly team meetings with documented minutes
- **Task Assignment:** Clear roles and responsibilities for each team member
- **Progress Tracking:** Use project management tools (Trello, Asana, GitHub Projects)
- **Documentation:** Keep detailed notes on decisions, challenges, and solutions

6.2 Industry Engagement

- **Professional Communication:** Formal emails and courteous interactions with company stakeholders
- **Respect for Confidentiality:** Obtain permission before using company data or discussing sensitive information
- **Regular Updates:** Inform company partners of progress and timeline
- **Gratitude:** Thank company representatives upon project completion

6.3 Technical Best Practices

- **Version Control:** Commit frequently with meaningful commit messages
- **Testing:** Implement unit tests or validation procedures
- **Documentation:** Write clear README files and inline code comments
- **Reproducibility:** Ensure anyone can run your code with clear instructions
- **Ethical Considerations:** Address bias, fairness, privacy, and security concerns explicitly

6.4 Academic Integrity

- **Original Work:** All code and analysis must be original (cite external sources properly)
- **Plagiarism Prevention:** Use proper citations for literature and acknowledge external libraries/tools
- **Collaboration:** Clearly distinguish individual and collaborative contributions
- **Honesty:** Report limitations and challenges transparently

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7. Support and Resources

7.1 Course Resources

- **Lectures and Tutorials:** Recordings, slides, and example code on LMS
- **Office Hours:** Weekly sessions with Dr. Milan Dordevic, Dr. Abdullah El Nokiti, Ms. Asma Damankesh
- **Discussion Forum:** Peer support and instructor clarifications

7.2 Technical Resources

- **Python Libraries:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy
- **Data Sources:** Kaggle, UCI Machine Learning Repository, company data (with permission)
- **Development Tools:** Jupyter Notebooks, VS Code, PyCharm, Git/GitHub
- **Documentation:** Official library docs, Stack Overflow, online tutorials

7.3 Writing and Presentation Support

- **Writing Center:** UOWD academic writing support
- **Presentation Workshop:** Optional guidance on effective presentation techniques
- **Peer Review:** Encourage teammates and classmates to review work before submission

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8. Frequently Asked Questions (FAQ)

Q1: Can I choose a fictional company?
A: No. The project explicitly requires identification of a real UAE-based company. Research-based approaches studying actual industry problems are acceptable, but the problem context must be realistic.

Q2: Can I use a dataset from Kaggle instead of company data?
A: Only if you clearly justify that the public dataset authentically represents the company's problem context. Preference is given to real company data or synthetic data modeled after real scenarios.

Q3: What if the company refuses to meet with us?
A: You may proceed with research-based analysis of generic industry problems. Document your research approach and discuss the limitations of not having direct company input.

Q4: How should we divide work in a team?
A: Clearly assign roles (project manager, data engineer, model developer, documentation lead, presenter, etc.). Document contributions and ensure balanced effort. All team members should understand the entire project.

Q5: Can we use pre-trained models (transfer learning)?
A: Yes, but you must justify the choice, explain fine-tuning or adaptation, and clearly document what is your original contribution versus what is pre-trained.

Q6: Is plagiarism detection software used?
A: Yes. Reports and code are checked for originality using Turnitin and code similarity tools. Proper citations and attributions are essential.

Q7: Can we change our project midway?
A: Only with instructor approval. If significant issues arise, contact Dr. Milan Dordevic, Dr. Abdullah El Nokiti, Ms. Asma Damankesh, Ms. Syama Kurungodathil as soon as possible to discuss modifications.

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9. Conclusion

The CSCI323 Modern Artificial Intelligence project is an opportunity to apply course concepts to authentic business problems, develop practical AI implementation skills, and engage with industry stakeholders. Success requires careful problem selection, rigorous technical execution, critical evaluation, and professional communication. By completing this project successfully, you will demonstrate not only technical competency but also the ability to translate complex AI concepts into tangible business value.

Questions? Contact Dr. Milan Dordevic, Dr. Abdullah El Nokiti, Ms. Asma Damankesh, Ms. Syama Kurungodathil during office hours or via email.

Good luck with your project!